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Mechanical Engineering

Metallic Materials and Technology 2025-26

Classification and Designation of Steels – EN and AISI Standard



Lecture 7

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I. Classification

Steels can be classified using different criteria:

- **by alloying element content**
- **by heat treatment**: steels for tempering, carburizing, nitriding, etc.
- **by applications**: structural, construction, tooling, etc.
- **by degree of workability**: “automatic”, cold forming, etc.

Typically standards do not follow only one criterion

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I. Classification- UNI EN 10020

If at least 1 element content > limits, the steel is named as **alloyed**, otherwise is said **unalloyed**

Element	Limit, % (in mass)
Mn	1,65
Cr	0,3
Ni	0,3
Mo	0,06
W	0,3
V	0,1

Unalloyed steels (plain carbon steels):

- “quality”
- “special” -> with specifications: i.e. CVN, hardenability, purity (bounded inclusion content)

Alloyed steels:

- “quality”: weldable with small grain size, for railways, for sheet being cold formed with deep drawing ability, for magnetic use, etc.
- “special” -> olters: tempering steels or carburizing, for tools, rapid steels, etc.

Stainless steels: Cr > 10,5%, C < 1,2%

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II. Alphanumeric designation

- Based on mechanical/physical and uses
- Based on chemical composition

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II. Alphanumeric Designation

A. Based on mechanical/physical properties and uses

[use, group] ([subgroup]) [property value**] ([additional symbols**])

Example:

S235JR

=

Structural use, R_{p02} 235 MPa, Impact energy 27 J

* subgroup, seldom used

** o arbitrary number

*** default

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II. Alphanumeric Designation

A. Based on mechanical/physical properties and uses

Use groups (a few):

S, structural

P, high pressure

R, railway

D, deep drawing sheets

H, high strength deep drawing sheets

M, magnetic properties

...

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II. Alphanumeric designation

A. Based on mechanical/physical properties and applications

Group S: Structural steels

Weldable beams, **UNI-EN-10025**

old	Prsnt	R _{p02} [MPa]	R _m [MPa]	ε _{min} [%]	C	Mn	Si	S,P
Fe 360	S235	235	360-510	20	<0,17	<1,4	-	<0,035
Fe 510	S355	355	510-680	15	<0,2	<1,6	<0,55	<0,035

S235	Diameter [mm]	<100	100-150	150-250
	R _m [MPa]	360-510	350-500	340-490

- ✓The yield stress and ultimate stress are prescribed and can be provided by the producer by various combinations of compositions and thermomechanical treatments
- ✓The max limits for C, Mn are prescribed for welding purposes

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II. Alphanumeric designation

A. Based on mechanical/physical properties and applications

Group S: Structural steels

Additional symbols

Examples:

S235JR: resilience at least 27 J at temperature of +20 °C

S235J2: resilience at least 27 J at temperature of -20 °C

In general: S235XY

X is one of:

J: > 27 J

K: >40 J

L: >60 J

Y is one of:

R: 20 °C (Room temperature)

0: 0 °C

2: -20 °C

3: -30 °C

...

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II. Alphanumeric designation

A. Based on mechanical/physical properties and applications

Group D: deep drawing sheets

Subgroup (*D = Drawing*):

- **DX** – not specified
- **DC** – cold rolled
- **DD** – hot rolled- **UNI-EN-10111:**

old	present	R _{p02} [MPa]	R _m [MPa]	A [%]	C	Mn
Fe P11	DD11	170-340	440	29	<0,12	<0,6
Fe P13	DD13	170-310	400	29	<0,08	<0,4

(11, 13: conventional numbers)

- ✓ The yield stress and ultimate stress are prescribed and can be provided by the producer by various combinations of compositions and thermomechanical treatments
- ✓ The max limits for C, Mn are prescribed for welding purposes

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II. Alphanumeric designation

A. Based on mechanical/physical properties and applications

Group H: high strength deep drawing sheets

UNI-EN-10346

group	subgroup Rolling type	property R _m [MPa]	Additional symbols Microstructure
H	C - cold	T(n)nnn	X - dual phase
			F - ferritic-bainitic
	D - hot		T - TRIP
			C - complex phase
			M - martensitic

grade	R _{p02} [MPa]	R _m [MPa]	A [%]	C	Mn
HCT600X	340 - 420	> 600	> 20	< 0.17	< 2.2
HD T950C	720 - 920	> 950	> 9	< 0.25	< 2.2
HDT1200M	900 - 1150	> 1200	> 5	< 0.25	< 2

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II. Alphanumeric designation

B. Based on chemical composition (1/2)

Unalloyed steels with Mn < 1%:

CNN

NN = [nominal % C] x 100

Es.: **C45**, **C10**

Unalloyed steels with Mn > 1% or alloying elements with each < 5% (*low alloy steels*)

NN [El₁][El₂]...[El_j] [n₁]-[n₂]-...[n_j]

NN = [nominal % C] x 100

El_i : symbols of alloy elements

n_i : [nominal % element] x factor

NOT all elements appear in the grade notation
i.e., Mn usually is not prescribed when < 1%

Ex.: **28Mn6**, **42CrMoS4**, **20NiCrMo2**

Elements	factor
Cr, Ni, Mn, Co, Si, W	4
Al, Mo, ...	10
S, ...	100
B	1000

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II. Alphanumeric designation

B. Based on chemical composition (2/2)

Alloyed with at least one element with > 5% (except for rapid steels) :

X NN [El₁][El₂]...[El_j] [%₁]-[%₂]-...[%_j]

NN = [nominal % C] x 100

El_i : Symbols of alloying elements

%_i : nominal alloy element content

Ex.: **X100CrMoV5**, **X38CrMoV5-3**

Rapid steels (High speed Steels, HS)

HS [% W]-[% Mo]-[% V]-[% Co]

% ... : nominal content of the element,

Es.: **HS 6-5-3-8**

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II. Alphanumeric designation: EXAMPLES

B. Based on chemical composition (2/2)

Tempering steels – UNI EN 10083

	S	C	Mn	Cr	Mo	P
C45E	<0,035	0,42-0,5	0,5-0,8	-	-	<0,035
C45R	0,02-0,8					
42CrMo4	<0,035	0,38-0,45	0,6-0,9	0,9-1,2	0,15-0,3	<0,035
42CrMoS4	0,02-0,8					

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II. Alphanumeric designation : EXAMPLES

B. Based on chemical composition (2/2)

Carburizing steels – UNI EN 10084

	S	C	Mn	Cr	Ni	Mo
C10E	<0,035	0,07-0,13	0,3-0,6	-	-	-
C10R	0,02-0,04					
20NiCrMo2-2	<0,035	0,17-0,23	0,65-0,95	0,35-0,7	0,4-0,7	0,15-0,25
20NiCrMoS2-2	0,02-0,04					

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II. Alphanumeric designation : EXAMPLES

B. Based on chemical composition (2/2)

Tool steels – UNI EN ISO 4957

	C	Mn	Cr	Mo	V	Si
X100CrMoV5	0,95–1,05	0,2–0,6	4,8-5,5	0,9-1,2	0,15-0,35	0,1-0,4
X38CrMoV5-3	0,35-0,4	0,3–0,5	2,7-3,2	2,7-3,2	0,4–0,6	0,3–0,5

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IV. American designation AISI / SAE

Example: tempering steels

[group][%C x 100]
AISI 1040 or AISI 4340

EN approximate correspondence

	Number	% C	% Mn	% Si	% Ni	% Cr	Others
C45E →	1020	0.18–0.23	0.30–0.60				
	1040	0.37–0.44	0.60–0.90				
	1060	0.55–0.65	0.60–0.90				
	1080	0.75–0.88	0.60–0.90				
	1095	0.90–1.03	0.30–0.50				
42CrMo4 →	1140	0.37–0.44	0.70–1.00				0.08–0.13% S
	4140	0.38–0.43	0.75–1.00	0.15–0.30		0.80–1.10	0.15–0.25% Mo
38NiCrMo4 →	4340	0.38–0.43	0.60–0.80	0.15–0.30	1.65–2.00	0.70–0.90	0.20–0.300% Mo
	4620	0.17–0.22	0.45–0.65	0.15–0.30	1.65–2.00		0.20–0.30% Mo

Low alloy steels

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AISI/SAE Table & Alloying Elements effects



SAE-AISI	Classification
10XX plain carbon 11XX free machining S 12XX free machining S.P 13XX Mn	Plain carbon steels Low carbon steels: 0 - 0.25 wt.%C Medium carbon steels: 0.25 - 0.55 wt.%C High carbon steels: > 0.55 wt.%C
2XXX	Nickel alloy steel 5 %Ni, ↑ UTS without reducing toughness 8 - 12 %Ni, ↑ Impact energy at low T 15 - 25 %Ni (with Al, Cu and Co) , ↑ magnetic properties (Alnicometals) 25 - 35 % Ni, ↑ Rcorr at high T
3XXX	Ni-Cr alloy steels Highly tough and ductile steels, ↑ Rwear, ↑ H, ↑ Rcorr
40XX Mo 41XX Cr-Mo 43XX Ni-Cr-Mo	Mo alloy steels Strong carbide former, ↑ H, ↑ HRC at high T, ↑ UTS in low C steels; very important element to prevent temper embrittlement in medium low alloy steels.
51XX Cr	Cr alloy steel Very strong ferrite strengthener in low C steels; ↑ core toughness, ↑ Rwear in carburizing steels.
86XX, 87XX, 93XX 94XX, 97XX, 98XX	Ni, Cr, Mo, V, Si: ↑ UTS, ↑ specific strength , ↑ Rcorr.

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Exercises on Steel Nomenclature

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III. Numeric Designation – UNI EN 10027

1.XXYY

1 = "steel", XX = group, YY = progressive number.

"00" base steels

S253JR -> 1.0037

S355JR -> 1.0045

"03" steels with C < 0.12%

DD11 -> 1.0332

DD12 -> 1.0335

"09"

HCT600X -> 1.0941

HDT950C -> 1.0958

HDT1200M -> 1.0965

"11" e "12" Carbon steels

For structural/mechanical uses

C45E -> 1.1191

C45R -> 1.1201

C10E -> 1.1121

C10R -> 1.1207

"72" steels with Cr and Mo, Mo ≤ 0.35%

42CrMo4 -> 1.7225

42CrMoS4 -> 1.7227

"65" steels with Cr, Ni and Mo

20NiCrMo2-2 -> 1.6523

20NiCrMoS2-2 -> 1.6526

NOT TO BE ASKED AT THE EXAM

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AISI/SAE Steel Nomenclature

Carbon steels	10XX	Plain carbon, Mn 1.00% max	Last two digits indicate amount of C in hundredth
	11XX	Resulfurized free machining	
	12XX	Resulfurized/rephosphorized free machining	
	15XX	Plain carbon, Mn 1.00-1.65%	



Manganese: Improves hardenability, strength, abrasion resistance and machinability; deoxidizes the molten steel and reduces hot shortness; decreases weldability.

Manganese steel	13XX	Mn 1.75%
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Nickel – improve strength without loss of ductility
Enhances case hardenability

Nickel steels	23XX	Ni 3.50%
	25XX	Ni 5.00%

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AISI/SAE Steel Nomenclature

Nickel-chromium steels	31XX	Ni 1.25%, Cr 0.65-0.80%
	32XX	Ni 1.75%, Cr 1.07%
	33XX	Ni 3.50%, Cr 1.50-1.57%
	34XX	Ni 3.00%, Cr .77%

Nickel with Chromium: improved elastic limit, hardenability, impact resistance and fatigue resistance



Molybdenum steels	40XX	Mo 0.20-0.25%
	44XX	Mo 0.40-0.52%

Molybdenum: Improves hardenability, wear resistance, toughness, elevated temperature strength, creep resistance and hardness; *minimizes temper embrittlement.*



Chromium-molybdenum steels	41XX	Cr .50-.95%, Mo .12-.30%
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Nickel-chromium-molybdenum steels	43XX	Ni 1.82%, Cr 0.50-0.80%, Mo 0.25%
	47XX	Ni 1.05%, Cr 0.45%, Mo 0.20-0.35%

Molybdenum with nickel and/or chromium – adds hardness, reduces brittleness, increase toughness